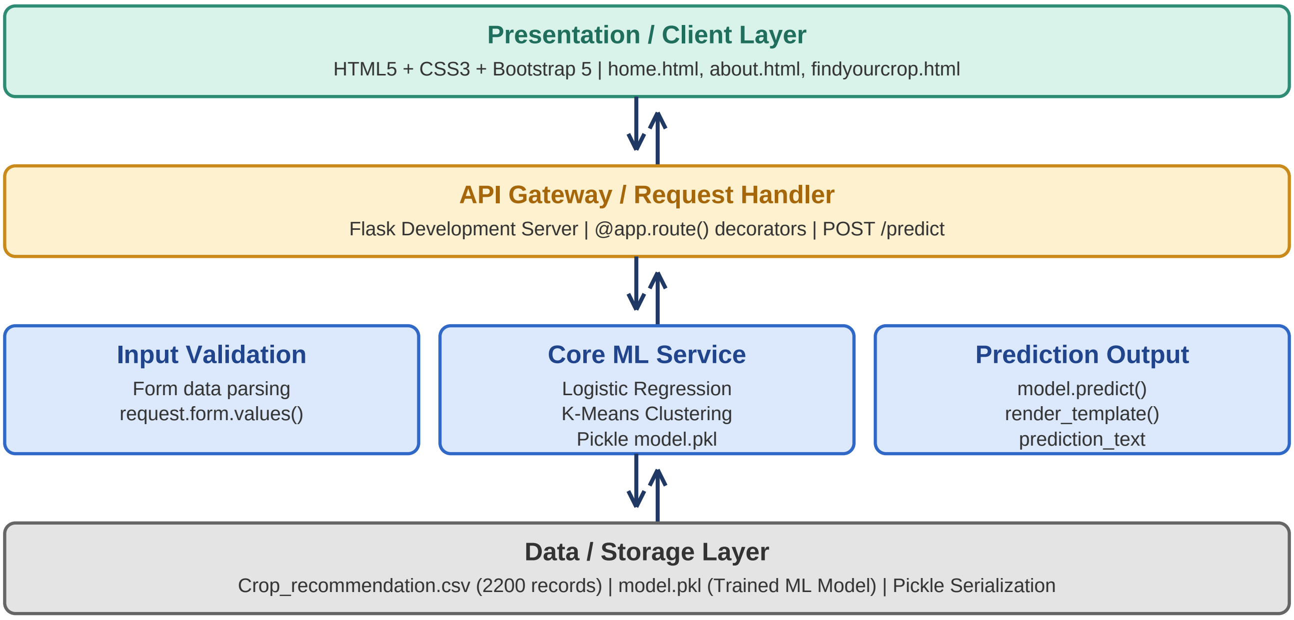
Solution Architecture

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below provides a high-level visual representation of the OptiCrop solution architecture, mapped across Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user-facing web interface where farmers input soil and climate parameters and view crop recommendation results.	HTML5, CSS3, Bootstrap 5, Jinja2 Templates
API Gateway / Request Handler	Flask development server manages all HTTP routing, handles POST requests from the prediction form, and renders response templates.	Flask, @app.route(), render_template()
Core ML Service	Processes validated input data, loads the trained ML model from pickle file, and generates intelligent crop predictions using Logistic Regression.	Scikit-learn, Logistic Regression, K-Means, Pickle
Data / Storage Layer	Stores the agricultural dataset (CSV) used for model training and the serialized trained ML model (.pkl) for reuse during prediction.	Crop_recommendation.csv, model.pkl, Pickle, Pandas